

Building a PC Control System Using Wonderware InTouch SCADA and Allen Bradley PLC



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Industrial control systems are always very complex and cumbersome to handle. For electronics engineers, it is becoming very essential to understand the computer control system using a programmable logic controller (PLC) and supervisory control and data acquisition (SCADA) system.

Here is a very efficient and expandable robust control system solution for industrial purposes. For the proposed system, we have used AB Allen Bradley MicroLogix 1400 PLC 1766, Wonderware InTouch 2012R2 SCADA system and KEPServer version 5 software for communicating between the PLC and SCADA. Wonderware InTouch SCADA is becoming increasingly popular due to its powerful user-friendly graphics as shown in Fig. 1.

The entire process can be summarised in the following steps:

1. The RSLogix family of IEC-1131-compliant Ladder logic programming

packages help you maximise performance, save project development time and improve productivity. This family of products has been developed to operate on Microsoft Windows operating systems. Supporting Allen Bradley SLC 500 and MicroLogix families of processors, RSLogix 500 was the first PLC programming software to offer unbeatable productivity with an industry-leading user interface. PLC is programmed using Ladder programming in RSLogix software. Different inputs and outputs have different addresses. Mostly the field devices, which are to be controlled, are connected to the PLC via digital input, analogue input, digital output and RS232, among others. The address of each device, module and register is specific to each PLC manufacturer, including Allen Bradley, Siemens, Schneider, Mitsubishi, Hitachi and Delta.

2. The PLC is connected to the PC via the Ethernet. Since both PLC and SCADA are from different manufacturers, we used an intermediate OPC link (communicator) between these. This job is done by KEPServer, which is installed on the PC.

3. The PLC IP address is configured using software tool BOOTPServer.

4. When the main SCADA is running, applications like KEPServer should be running in the background and PLC should be in online mode.

5. Using SCADA, a graphical user interface (GUI) is built to monitor and control various parameters like current/voltage and to switch on/off various devices.

Selection of Allen Bradley PLC by Rockwell Automation

For a beginner, it is better to start with the MicroLogix family of PLCs from Allen Bradley. MicroLogix 1000 controller offers

Fig. 1: A typical Wonderware InTouch graphical user interface

